4a. bentuk kutub

diketahui : $z_1 = (a+2) + (b+1)i = 4+7i$

$z_2 = (b+1) - (a+2)i = 7-4i$

$z_1$ $r_1 = \sqrt{(4^2+7^2)} = \sqrt{65}$

$\theta = \arctan(7/4) = 60.2551^\circ$

$= 1.0517$ rad

$z_1 = \sqrt{65}\ \text{cis}\ (60.2551^\circ)$

$z_2 = r_2 = \sqrt{(7^2+(-4)^2)} = \sqrt{65}$

$\theta = \arctan(-4/7) = 29.7449^\circ$

$= -0.5191$ rad

$z_2 = \sqrt{65}\ \text{cis}\ (-29.7449^\circ)$

4b. operasi kompleks

1. penjumlahan

$z_1 + z_2 = (4+7i) + (7-4i) = 11+3i$

2. perkalian

$z_1 z_2 = (4+7i)(7-4i)$

$= 28 - 16i + 49i - 28i = 56+33i$

3. pembagian

$z_1/z_2 = (4+7i)/(7-4i) = (28+16i+49i+28i^2)/65$

$= (28+65i-28)/65 = i$

$\times (7+4i)/(7+4i)$

$= (4+7i)(7+4i)/(65)$

$=$

$= |z_1 - z_2|$

$z_1 - z_2 = (4+7i) - (7-4i) = -3+11i$

$|z_1 - z_2| = \sqrt{((-3)^2+11^2)} = \sqrt{130}$

deli得力

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4c. de moivre

$z_1 = \sqrt{65}\ cis(\theta)$ $= 65^2\ cis(241.0204^\circ)$

$(z_1)^4 = (\sqrt{65})^4\ cis(4\theta)$ $= 4225\ cis(241.0204^\circ)$

Jika dihitung al jabar (dua 4 4) hasil sama

4d akar rubik

$w = 8\ cis(90^\circ)$ rumus : $\theta_k = (90^\circ + 360^\circ k)/3$

$r = 8 \rightarrow$ akar : 2 $k = 0 \rightarrow \theta = 30^\circ$ $k = 1 \rightarrow \theta = 150^\circ$

$k = 2 \rightarrow \theta = 270^\circ$

4f interpretasi

$\arg(z_1/z_2)$ : selisih fase penjadian

artinya = beda timing puncak penjadian dan far produk

Fourier penting krn : memecah pola musiman, deteksi tren & siklus penjualan

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